AARON DANIEL

a.sterling.daniel@gmail.com • 512.906.6613 • linkedin.com/in/aaron-s-daniel

Data scientist and AI researcher with 6+ years experience in aerospace & defense, applying AI/ML to solve complex engineering challenges

EXPERIENCE

Senior Data Scientist, Lockheed Martin Missiles and Fire Control **2022 – 2026**

- Analytics Chatbot: Lead developer of tool for Production Operations, enabling natural language queries across enterprise datasets
- Wire Crimp Inspection Tool: Developed Inspection tool enabling Automated Optical Inspection on wire crimps on the production floor across multiple sites
- Recommendation Data Model and Analytics Suite: Recommendation model (SAP HANA, SQL) enabling data-driven decision making and resulting in $1.2M annual labor savings
- OCR, NLP for business automation: Developed front-end (Python, Streamlit) and back-end (Python, SAP HANA, Windchill) application which leverages OCR, NLP, and recommendation system to extract data from engineering drawings and recommend quality notes, resulting in $175K annual labor savings
- Analytics Development and Maintenance: Help maintain 40+ dashboards and data sources (SAP HANA / SQL) on Tableau Server aligned to business objectives and digital transformation with 200K+ user views annually

Aeronautical Engineer / Data Analyst, Lockheed Martin Aeronautics **2020 – 2022**

- Developed ML models for the identification and classification of aircraft maneuvers in flight test data
- Drove characterization of aircraft system performance in flight and lab test environments through manipulation, analysis, and visualization of large time series data sets with Python and Tableau

EDUCATION

University of Texas at San Antonio, MS Artificial Intelligence **2022 – 2025**

- Concentration: Computer Science, GPA 3.9/4.0
- Masters Thesis: "CROSS-MODALITY EVALUATION OF EXPLAINABLE AI METHODS ON IMAGE AND AUDIO CLASSIFICATION TASKS"
- Evaluation of six post-hoc XAI methods (Saliency, Integrated Gradients, Guided GradCAM, DeepLIFT, Input × Gradient, Occlusion) applied to both image (MNIST) and audio (AudioMNIST) classification tasks using a consistent CNN architecture with grayscale inputs (MNIST Images, AudioMNIST Spectrograms)
- Relevant Coursework: Advanced Machine Learning, Computer Vision, Deep Learning, Applications of AI Model Explainability (Independent Study)

Texas A&M University, BS Aerospace Engineering, Spanish Minor **2015 – 2020**

SKILLS

- Languages & Frameworks: Python (NumPy, Pandas, Scikit-learn, SciPy, OpenCV), SQL, TensorFlow, PyTorch
- AI/ML: Deep Learning, Computer Vision, NLP, GenAI, Explainable AI, Reinforcement Learning
- Technical Data Analysis: Multi-reference Frame Kinematics, Telemetry, Radar, Electronic Warfare, Time Series, Audio Data in both lab and flight test environments
- Tools: SAP HANA, Tableau, Streamlit, Git, Docker, OpenShift, AWS EC2